# Daromad

## Income evidence for borrowers who have no payslip

**Mode: Partner product — MKBANK, micro & family-business credit department**

**Team TYS** three members — technical lead · research & data · compliance & business case

**Live prototype** https://tys-fintech.vercel.app **Repository** https://github.com/hmunisbekk-droid/TYS-fintech

FinTech / AI in Finance · partners MKBANK and the Central Bank of Uzbekistan · final defence, October 2026

# The regulation names the evidence — the self-employed have exactly one line

**CBU regulation on the debt-burden ratio, cl.9 — income a lender may rely on**

- **Employer-confirmed salary** — no employer — the borrower is the business
- **Pension** — working-age applicant, not a pensioner
- **Funded-pension contributions** — none: contributions come through an employer
- ✓ **Income credited to the borrower's bank accounts** — the only line a self-employed borrower has
- **Taxes paid** — self-employed income is untaxed; no return is filed
- **Interest, dividends, rental income** — rarely applicable; easy to document when it is

## 50% cap

ПДН limit for all loans since 1 Jan 2025 (cl.10); 12-month arithmetic mean of income (cl.7)

## ~140,000

micro-projects MKBANK targets in 2026 (47,975 in 2025); 1,062 mahalla agents

## 60–90 min

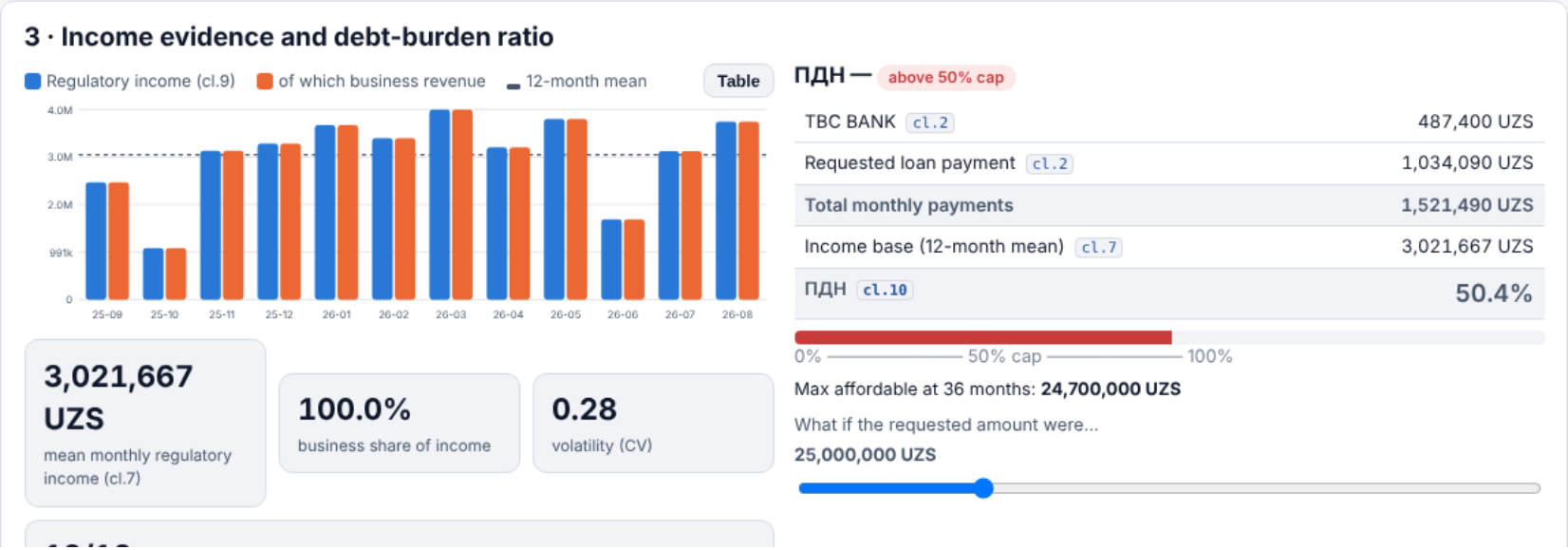
an officer's manual reading of 12 months of statements per application (our estimate; pilot measures it)

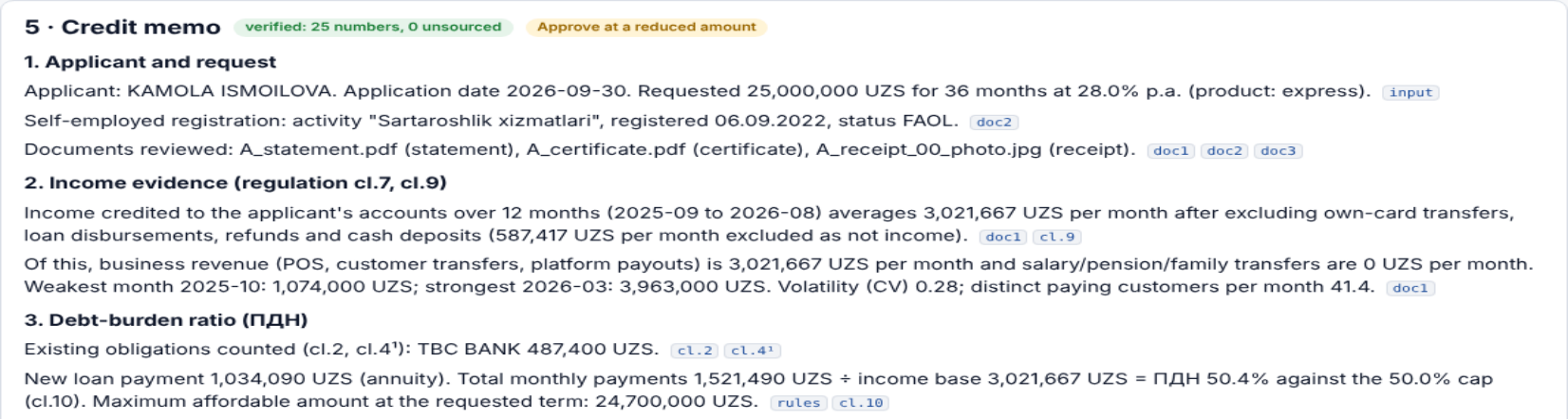
## 0 lines

of official income records for ~790k registered self-employed (untaxed, unfiled, no cash register required)

Sources: lex.uz/docs/4654149 and 6809992 (CBU regulation and 2024 amendments) · spot.uz 27 Mar 2026 (MKBANK, bank's own figures) · yuz.uz (self-employed registrations, 2021) · buxgalter.uz

# What the officer gets in under five seconds




## 12-month regulatory income

cl.7 mean, cl.9 definition; own-card transfers, loan proceeds, refunds and cash deposits excluded

## ПДН with citations

36-month spreading, guarantor rule, family-programme exemption, lender-type rules (banks: no small-loan exclusion, 15% quota is portfolio-level), 50% cap, maximum affordable amount

## Integrity report

balance chain, duplicates, coverage, gaps, pre-application spikes, circular transfers, holder match

## Verified memo

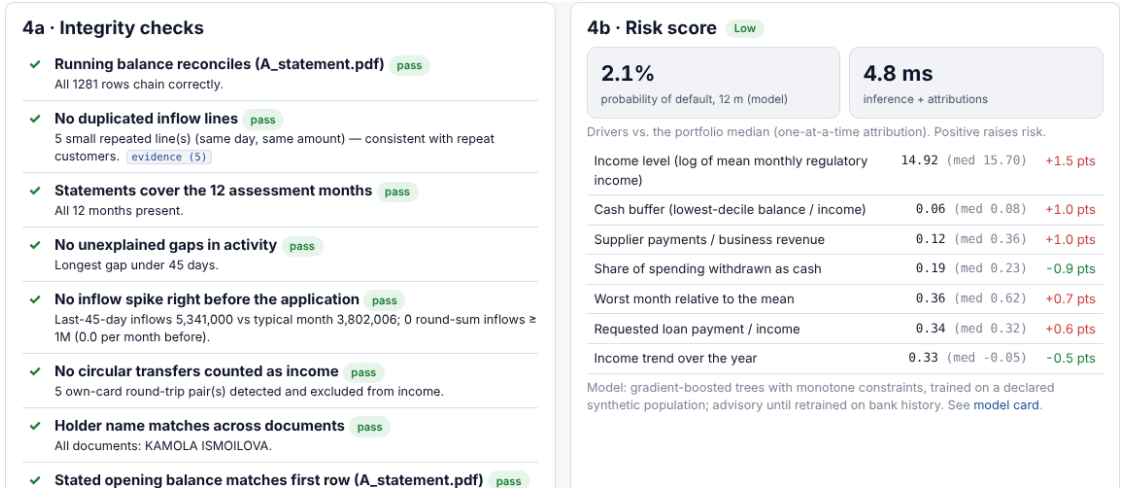
every number traces to its source — statement lines (click to open them), a clause, or the model; the verifier refuses unsourced numbers; hash-chained audit log

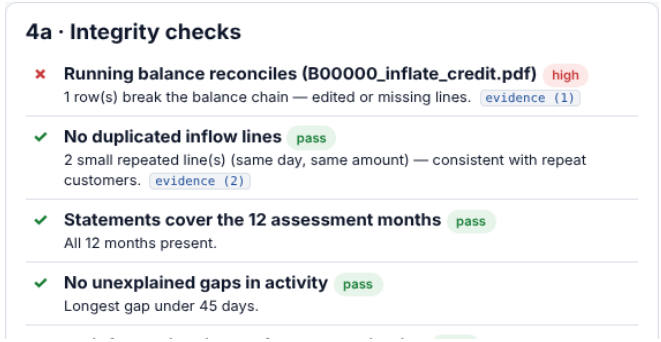
# Live demo — on our samples and on anything you upload

Open on your laptop or phone

**https://tys-fintech.vercel.app**

1. Load sample A — statement PDF, Soliq certificate, receipt photo parse in the tab (no server, no API)
2. Income series, ПДН with clause citations, maximum affordable amount; drag the slider to cross the 50% cap
3. Integrity checks and risk score with drivers; click any evidence link to see the statement line
4. Memo: verifier badge shows how many numbers were checked and that none is unsourced
5. Record the decision; export the case with its hash-chained audit log
6. Then: Load tampered statement — the balance chain breaks at the inflated inflow; upload your own documents




Scan to open the live prototype on your phone (New case → Load sample A)

# Seven components, one optional language model, zero API keys

### 1 · Intake

pdf.js with coordinates; XLSX/CSV; tesseract.js OCR uzb+rus+eng, self-hosted; field confidence, tap-to-confirm

### 2 · Categoriser

multilingual rules + counterparty patterns (recurring payer → family; same mask both ways → own card)

### 3 · Cash-flow

12-month series; 18 features identical in TypeScript and Python (parity test)

### 4 · Rule engine

ПДН clause by clause: 36/180-month spreading, guarantor rule, lender-type exclusions, family-programme exemption, 50% cap, back-solved amount

### 5 · Risk model

XGBoost, monotone constraints, JSON trees evaluated in the browser; attributions vs portfolio median

### 6 · Integrity

balance chain, duplicates, coverage, gaps, round-sum spikes, circular transfers, holder match

### 7 · Memo + audit

template from a fact table; verifier rejects unsourced numbers; SHA-256 hash-chained log

### Perimeter

Prototype: static site, every byte of a document stays in the browser. Production: the same bundle on the bank's server, SSO, audit log in the bank DB, core-banking pull for own customers. LLM optional, narrative only, behind the same verifier.

**Speed: statement 1.2–1.8 s · receipt OCR 1.6 s · everything else < 50 ms · marginal cost per application: zero**

Human in the loop: the engine recommends and computes the maximum affordable amount; the officer decides; an override needs a reason, written into the chain.

# Measured through the real app, not typed into a slide

**60/60**

generator PDFs: exact rows, balance chain reconciled (2 layouts, RU/UZ)

**8/8**

unseen layouts exact — CSV/XLSX/TSV variants incl. a wallet export (list on the model card)

**40/40**

tampered statements caught, 4 tamper types; 0 false alarms on clean ones

**95%**

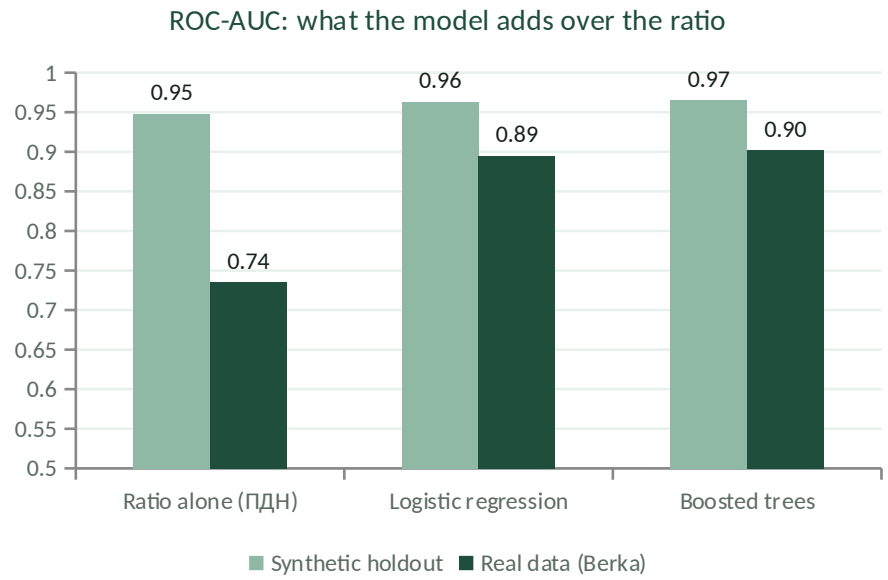
receipt OCR total exact, 60 camera-augmented photos; INN 100%, date 97%

**0.965**

ROC-AUC, synthetic holdout n=500; logistic 0.963, ratio alone 0.948

**0.902**

ROC-AUC, real bank data (Berka, 443 loans, 44 bad; 95% CI ≈ 0.86–0.94); ratio alone 0.735




Honest reading: on synthetic data the ratio already carries most of the signal by construction; on real statements the cash-flow features add +0.17 AUC over the ratio, and logistic regression comes close to the trees — we ship the trees for the monotone constraints and non-linear thresholds, not for the last 0.01. Categoriser: 99.96% on 10,481 lines of the generator's own vocabulary — will fall on real data. Memo verifier: 0 unsourced numbers across a 300-scenario sweep (amounts × terms × products × borrowers), enforced at run time.




All numbers are read from JSON produced by ml/train.py, ml/berka_validate.py, npm test and tests/*_eval.py (see the Model card page).

**Caveat on the slide, not in the footnote: the risk score is advisory until retrained on the bank's own history inside the perimeter — that is the pilot's first deliverable.**

# No partner data, so we built a declared generator

1

## Archetypes

10 trades (tailor, taxi, dehkan farmer, grocer, reseller, tutor…) with declared seasonality, customer counts, channel mix, expense structure

2

## Simulation

2,000 borrowers · 5,689,589 transactions · existing loans with annuity schedules · shocks · default label from a latent affordability score (10.5% rate)

3

## Rendering

60 statement sets as PDF (text layer) + XLSX + CSV in RU and UZ layouts; 180 fiscal receipts; Soliq-style certificates with QR

4

## Camera model

perspective, rotation, lighting gradient, blur, sensor noise, JPEG — then measured through the same OCR the officer uses

5

## Tampering

40 statements: inflated inflow, deleted debit, duplicated row, window dressing — labelled for the integrity evaluation



**HUMO CARD STATEMENT**
Выписка по карте

Владелец: KAMOLA ISMOILOV
Карта: 5614 **** **** 4369
Период: 2025-09-01 — 2026-08-31
Валюта: UZS

Входящий остаток: **668 582.52**

| Дата | Время | Описание операции | Расход | Приход | Остаток |
|---|---|---|---|---|---|
| 2025-09-01 | 07:33 | P2P HUMO 8600 **** 9613 | | 50 000.00 | 718 582.52 |
| 2025-09-01 | 08:33 | IJARA TO'LOVI P2P 8600 **** 2210 | 674 000.00 | | 44 582.52 |
| 2025-09-01 | 08:54 | NAQD PUL KIRITISH BANKOMAT | | 350 000.00 | 394 582.52 |
| 2025-09-01 | 09:03 | HUDUDGAZTAMINOT KOMMUNAL | 131 000.00 | | 263 582.52 |
| 2025-09-01 | 14:57 | HUDUDIY ELEKTR TARMOQLARI KOMMUNAL | 90 000.00 | | 173 582.52 |
| 2025-09-01 | 16:24 | UCELL ABONENT | 62 000.00 | | 111 582.52 |
| 2025-09-02 | 14:09 | BARAKA MARKET TOSHKENT | 103 000.00 | | 8 582.52 |
| 2025-09-03 | 16:36 | POS KAMOLA SARTAROSHLIK TERMINAL | | 52 000.00 | 60 582.52 |
| 2025-09-04 | 07:29 | P2P HUMO 9860 **** 9719 | | 17 000.00 | 77 582.52 |
| 2025-09-04 | 12:45 | POS KAMOLA SARTAROSHLIK TERMINAL | | 40 000.00 | 117 582.52 |
| 2025-09-04 | 13:46 | POS KAMOLA SARTAROSHLIK TERMINAL | | 22 000.00 | 139 582.52 |
| 2025-09-04 | 16:24 | P2P HUMO 8600 **** 6177 | | 38 000.00 | 177 582.52 |
| 2025-09-04 | 16:48 | P2P HUMO 9860 **** 4617 | | 39 000.00 | 216 582.52 |
| 2025-09-04 | 17:40 | POS KAMOLA SARTAROSHLIK TERMINAL | | 27 000.00 | 243 582.52 |
| 2025-09-04 | 22:21 | ANGLESEY TOSHKENT | 63 000.00 | | 180 582.52 |
| 2025-09-05 | 20:06 | POS KAMOLA SARTAROSHLIK TERMINAL | | 32 000.00 | 212 582.52 |
| 2025-09-05 | 20:29 | P2P HUMO 8600 **** 9240 | | 31 000.00 | 243 582.52 |
| 2025-09-05 | 22:57 | OPTOM SAVDO MCHJ TO'LOV | 104 000.00 | | 139 582.52 |
| 2025-09-06 | 14:53 | POS KAMOLA SARTAROSHLIK TERMINAL | | 64 000.00 | 203 582.52 |
| 2025-09-06 | 17:24 | HAVAS TOSHKENT | 131 000.00 | | 72 582.52 |
| 2025-09-07 | 11:40 | P2P HUMO 9860 **** 3683 | | 66 000.00 | 138 582.52 |
| 2025-09-07 | 13:08 | POS KAMOLA SARTAROSHLIK TERMINAL | | 31 000.00 | 169 582.52 |
| 2025-09-07 | 20:29 | OPTOM SAVDO MCHJ TO'LOV | 88 000.00 | | 81 582.52 |
| 2025-09-08 | 07:44 | POS KAMOLA SARTAROSHLIK TERMINAL | | 43 000.00 | 124 582.52 |
| 2025-09-08 | 15:53 | POS KAMOLA SARTAROSHLIK TERMINAL | | 39 000.00 | 163 582.52 |
| 2025-09-08 | 16:34 | OPTOM SAVDO MCHJ TO'LOV | 106 000.00 | | 57 582.52 |
| 2025-09-08 | 18:41 | KORZINKA TOSHKENT | 42 000.00 | | 15 582.52 |
| 2025-09-09 | 08:34 | POS KAMOLA SARTAROSHLIK TERMINAL | | 50 000.00 | 65 582.52 |
| 2025-09-09 | 09:50 | POS KAMOLA SARTAROSHLIK TERMINAL | | 27 000.00 | 92 582.52 |
| 2025-09-09 | 21:57 | HAVAS TOSHKENT | 50 000.00 | | 42 582.52 |
| 2025-09-10 | 12:30 | POS KAMOLA SARTAROSHLIK TERMINAL | | 30 000.00 | 72 582.52 |
| 2025-09-10 | 18:43 | POS KAMOLA SARTAROSHLIK TERMINAL | | 46 000.00 | 118 582.52 |
| 2025-09-10 | 22:08 | P2P HUMO 8600 **** 3739 | | 34 000.00 | 152 582.52 |
| 2025-09-11 | 10:18 | P2P HUMO 9860 **** 5631 | | 33 000.00 | 185 582.52 |
| 2025-09-11 | 17:02 | POS KAMOLA SARTAROSHLIK TERMINAL | | 37 000.00 | 222 582.52 |
| 2025-09-11 | 18:42 | P2P HUMO 8600 **** 7135 | | 44 000.00 | 266 582.52 |
| 2025-09-11 | 21:26 | P2P HUMO 9860 **** 6338 | | 30 000.00 | 296 582.52 |
| 2025-09-11 | 22:12 | P2P HUMO 9860 **** 6674 | | 34 000.00 | 330 582.52 |
| 2025-09-12 | 15:07 | POS KAMOLA SARTAROSHLIK TERMINAL | | 27 000.00 | 357 582.52 |
| 2025-09-12 | 18:20 | P2P HUMO 9860 **** 2803 | | 24 000.00 | 381 582.52 |



KAMOLA SARTAROSHLIK
Xorazm, Mustaqillik ko'chasi 9
STIR/INN: 342056057
FISKAL CHEK № 58332
01.03.2026 13:36
Kamola
Sana
Kassir
BOSH YUVISH 45 000
3 x 15 000
SOCH OLISH 40 000
1 x 40 000
JAMI 85 000
To'landi (UZCARD) 85 000
QQS 12% 9 107
FM: UZ118914450618 FP: 135878439006
ofd.soliq.uz
Xaridingiz uchun rahmat!

Synthetic statement (Humo-style layout, Russian labels) and a camera-augmented synthetic fiscal receipt. No real person is represented; names come from a short public list, card numbers are random and masked.

```
Regenerate:  python -m synth.cli --n 2000 --render 60 --
receipts 180 --tamper 40 --seed 42
```

Also: methodology check on real anonymised bank data (Berka, PKDD'99) — not redistributed, downloaded by script.

# Pilot: three months, five branches, fifty mahalla agents

## Cost to build and run the pilot (assumptions, UZS)

| Line | Assumption | 3-month cost |
|---|---|---|
| Team: 3 engineers/analysts (TYS) | 3 × 3 months × 15M | 135M |
| Bank credit analyst, part-time | 0.5 FTE × 3 months × 15M | 22.5M |
| Infrastructure | 1 VM in the bank DC, no GPU | ≈ 3M |
| Integration (core banking, bureau) | bank IT, 20 person-days | ≈ 30M |
| Data, licences, API spend | none — open-source, no external API | 0 |
| **Total pilot (≈ $15k)** | | **≈ 190M** |

Running cost per application: zero marginal (static bundle; no API, no GPU). Optional narrative LLM on one on-prem GPU: well under 1 US cent per application.

Timeline: month 1 shadow mode (engine runs beside the manual process; we measure minutes and agreement) · months 2–3 assisted mode (officer works from the memo) · success metrics: minutes per application, agreement with committee, override rate, ПДН audit findings, retrained model AUC on bank history.

## Value — the officer-hours formula (every term is a slider)

**90 → 25 min**

document handling per application, before → with Daromad (assumption 1; the pilot measures it)

**≈ 42,000 / yr**

addressable applications: 30% of MKBANK's 140,000 micro-projects (bank's own 2026 target) — those outside the ПДН-exempt family programme (assumption 2)

**≈ 45,000 h**

officer-hours saved per year = 42,000 × 65 min ≈ 23 FTE-years at 2,000 h

**≈ 4.1 bn UZS/yr**

23 FTE × 15M/month × 12 — officer time only; at a 10% addressable share ≈ 1.4 bn

**Payback:** the ≈ 190M pilot itself (5 of 181 branches, 3 months) does not pay back — it measures the two assumptions. After roll-out the saving of ≈ 4.1 bn/yr repays the pilot within its first month; at a 10% addressable share, within two months. Not counted: faster time-to-decision for the borrower, fewer ПДН compliance findings, better risk selection — claimed only after the pilot measures them.

Who pays: MKBANK (credit department / digital budget). Who owns: micro & family-business credit department; IT deploys; mahalla agents and loan officers use it.

# Hard to copy in a weekend — and what comes next

## The replication test — what the same LLM tools do not give you

- Statement parsers that learn columns from the data rows (two Uzbek layouts today, any text-layer PDF, XLSX, CSV) and reconcile the balance chain
- A categoriser lexicon in Uzbek Latin, Cyrillic and Russian with counterparty-pattern rules — and the pilot plan to grow it on bank data
- A rule engine mapped clause by clause to the live ПДН regulation, with its exemptions and the back-solved maximum amount
- A memo verifier that makes unsourced numbers impossible — zero hallucination by construction, no LLM needed
- A declared, calibrated synthetic population with rendered documents, a camera model and labelled tampering — a test bed the bank can keep using
- Evaluation through the real app, reproducible with one command each

## Next steps

- Week 1–2: scanned-statement OCR page by page; lexicon growth on anonymised bank descriptions; Uzbek/Russian memo templates
- Month 1: shadow mode in 5 branches; core-banking pull for MKBANK's own account holders; credit-bureau REST for obligations
- Month 2–3: assisted mode; retrain the risk model on bank history inside the perimeter; set risk bands with the bank's risk appetite; fairness review by region
- After the pilot: Central Bank conversation — the same engine computes ПДН uniformly for any lender, a supervisory check at zero marginal cost
- Product scope today: Ommabop 2, Express, Business Online 2 (ПДН applies) and family-programme loans (exempt from the cap, capacity still assessed)

# Daromad

Income evidence for borrowers who have no payslip — regulation-native, perimeter-native, verified.

**Live:** https://tys-fintech.vercel.app
**Repo:** https://github.com/hmunisbekk-droid/TYS-fintech
**Team TYS**

# Appendix A · ПДН rule engine, clause by clause

| Clause | Rule as implemented | Where it shows |
|---|---|---|
| **cl.1** | ПДН = average monthly payments on loans/microloans ÷ average monthly income (borrower + co-borrower) | formula panel |
| **cl.2** | Includes the application under review and all loan agreements with all lenders; co-borrower loans counted. Lender-type reliefs (2024): banks («банк томонидан…») — only a portfolio quota, up to 15% of the number of loans to individuals outside the requirement; MFOs and pawnshops («микромолия ташкилоти ва ломбард томонидан…») — excluded when amount + outstanding principal ≤ 50 BRV; MFOs lending to the self-employed — 75 BRV. Daromad defaults to bank: ratio always computed, quota shown as a note | lender-type selector, ПДН panel note |
| **cl.4** | Guarantor loans count only when payments are more than 30 days overdue («30 кундан ортиқ муддати ўтган тўловлари мавжуд бўлса») | obligations table, DPD column |
| **cl.4[1]** | Terms over 36 months: total payments spread over 36 («барча тўловлар 36 ойга тенг миқдорда тақсимланади»); mortgages over 180 months spread over 180 | per-line note |
| **cl.6** | Calculation date within 10 calendar days of the loan date | case date |
| **cl.7** | Average monthly income = arithmetic mean of the last 12 months; co-borrower income added | income base |
| **cl.9** | Evidence: employer salary data, pension, funded-pension contributions, income credited to bank accounts, taxes paid, interest/dividends/rent; other data allowed | memo section 2 |
| **cl.10** | Ratio ≤ 50% from 1 January 2025 (60% between 1 July 2024 and 1 January 2025) | cap, pass/fail |
| **cl.11** | Family-entrepreneurship programme loans exempt from the ratio requirement | exemption badge |

BRV = 440,000 UZS from 1 September 2026 (previously 412,000); 50 BRV = 22,000,000 UZS. Clause mapping checked against lex.uz/docs/4654149 and 6809992 on 10 Oct 2026; quotes verbatim in docs/REGULATION_QUOTES.md.

# Appendix B · Failure cases, perimeter, human in the loop

## Known failure cases (also on the Model card page)

- Scanned statements without a text layer: not OCR'd page by page yet — ask for the XLSX/CSV export
- One signed amount column and no balance column: signs taken from keywords only
- Merchant strings outside the lexicon (mixed scripts, transliteration) → default category, visible in the table
- Dark or heavily skewed receipt photos: OCR confidence < 60% → fields flagged for confirmation, never guessed
- Cash-only businesses: nothing to reconstruct — the memo says so; the officer attests
- Synthetic-to-real gap of the risk score until retrained inside the bank; risk bands are placeholders

## Perimeter, audit, human in the loop

- Prototype: static site; documents are parsed in the browser and never transmitted; nothing is stored server-side
- Production: same bundle on the bank's web server (Docker, no outbound network), SSO identity, audit log and cases in the bank DB
- Own customers' statements pulled from core banking; uploads only for other banks' statements and wallets
- Every analysis stamped with rule-engine, model and categoriser versions; hash-chained SHA-256 log; export with the case
- The engine recommends; the officer decides; overrides require a reason; committee above branch limits
- Personal data processed inside the bank with recorded consent; retention per bank policy; no gender or activity-type features in the model

# Appendix C · Data sources and licences

| Source | Used for | Licence / terms |
|---|---|---|
| Daromad synthetic data set (synth/, seed 42) | training, evaluation, demo samples | MIT, this repo; generator and parameters declared; no real person |
| Berka / PKDD'99 Financial data set (mirror github.com/jlacko/berka-dataset) | offline methodology check | academic challenge data, no licence stated; evaluation only, not redistributed |
| Tesseract tessdata_fast (uzb, uzb_cyrl, rus, eng); tesseract.js | receipt OCR | Apache 2.0 |
| pdf.js; SheetJS; Next.js; React | parsing, application | Apache 2.0 / MIT |
| XGBoost; scikit-learn; NumPy; pandas; OpenCV; Playwright; Jinja2; qrcode | training, rendering, evaluation | Apache 2.0 / BSD-3 / MIT |
| CBU regulation texts (lex.uz 4654149, 6809992) | rule engine, citations | public legal information |
| stat.uz wages; CBU statistics; MKBANK public pages (sponsored article, bank's own figures) | calibration anchors, business case | public information, cited |

Rules compliance: no hard-coded answers (every figure computed from uploaded documents at run time); no real personal or confidential data; all code written during the final by team TYS with LLM coding assistance (declared); deployment from the submitted repository; CI on every push.